

Risk Aversion and Portfolio Demand¹

For $t = 0, 1$ let u_t be a C^1 vN-M utility with positive first derivative everywhere. Consider

$$\alpha_t \in \arg \max_{\alpha \geq 0} \left\{ \int u_t(w + \alpha x) dF(x) \right\}.$$

where $\int x dF(x) > 0$, $F(x) > 0$ for some $x < 0$ (so F is nondegenerate) and F has bounded support.

Theorem Let u_0 be *strictly more concave* than u_1 : $u_0(z) = T(u_1(z))$ for all $z \in \mathbb{R}_+$, where $T : \text{Range}(u_1) \rightarrow \mathbb{R}$ is strictly concave. Then $\alpha_1 \geq \alpha_0$.

Proof: Wlog normalize u_0 so $T'(u_1(w)) = 1$. Define

$$g(\alpha, t) = \int u_t(w + \alpha x) dF(x),$$

and note that wlog we may take the domain of $g(\cdot, t)$ to be $\mathbb{R}_{++}$. (Why?) We have

$$\begin{aligned} g_\alpha(\alpha, 0) &= \int u'_0(w + \alpha x) x dF(x) = \\ &= \int T'(u_1(w + \alpha x)) u'_1(w + \alpha x) x dF(x) \pm T'(u_1(w)) \int u'_1(w + \alpha x) x dF(x) = \\ &= \int [T'(u_1(w + \alpha x)) - T'(u_1(w))] u'_1(w + \alpha x) x dF(x) + T'(u_1(w)) \int u'_1(w + \alpha x) x dF(x) < \\ &= T'(u_1(w)) \int u'_1(w + \alpha x) x dF(x) = T'(u_1(w)) g_\alpha(\alpha, 1) = g_\alpha(\alpha, 1) \end{aligned}$$

where the inequality follows since T' is strictly decreasing, F is nondegenerate, and $\alpha > 0$.² The last equality follows from the normalization $T'(u_1(w)) = 1$. The inequality gives us that, for all $\alpha > 0$, $g_\alpha(\alpha, 1) > g_\alpha(\alpha, 0)$ implying that g satisfies strictly increasing differences in (α, t) on $\mathbb{R}_{++} \times \{0, 1\}$, and so $\alpha_1 \geq \alpha_0$. (If you did not normalize $u'_0(w)$ to be 1 you would get $g_\alpha(\alpha, 1) > \gamma g_\alpha(\alpha, 0)$ where $\gamma = (T'(u_1(w)))^{-1} > 0$; now integrate both sides with respect to α in between α' and $\alpha'' > \alpha' > 0$ and use that inequality to show that g satisfies the SSCP in (α, t) .)

Exercise Redo the argument for the special case in which x takes on only two values: $x_1 < 0$ with probability π_1 and $x_2 > 0$ with probability π_2 and $\pi_1 x_1 + \pi_2 x_2 > 0$.

Remark 1 Note that $g_\alpha(\alpha_1, 1) = 0$. If $g(\cdot, 0)$ is concave, then the inequality $0 = g_\alpha(\alpha_1, 1) > g_\alpha(\alpha_1, 0)$ immediately implies $\alpha_0 \leq \alpha_1$ (otherwise the K-T condition for $t = 0$ would fail).

Remark 2 If $u'(w + \alpha x)x$ were strictly increasing in x , then the Theorem would follow from the *Covariance Inequality* (see my notes on this topic).

Remark 3 The Pratt Theorem is a special case of a more useful result. For u_t , $t = 0, 1$ satisfying the hypotheses of the Pratt Theorem if $\int u_1(z) dG(z) \geq \int u_1(z) dF(z)$ for cdf's $G \neq F$ on $\mathbb{R}_{++}$, where G crosses F exactly once from below,³ then $\int u_0(z) dG(z) > \int u_0(z) dF(z)$.

¹ John Pratt, "Risk Aversion in the Small and in the Large," *Econometrica* 32 (1964): 122-36.

² Since T' is strictly decreasing and $\alpha > 0$, $[T'(u_1(w + \alpha x)) - T'(u_1(w))]x < 0$ for all $x \neq 0$.

³ There is a $\hat{z} > 0$ such that $G(z) > F(z)$ if and only if $z > \hat{z}$. Note that if G and F have the same mean, then F is *riskier* than G .